CYBR 345 Test1 Spring 2026

Cryptography Clue Hunt: The Lost Access Credentials

Important:

- Do not delete or alter any text or questions in this document.
- Answer all questions directly below each prompt and show all required work clearly.
- Do not provide or seek help from others while completing this assignment. All work must be your own.

Scenario:

Students in your class formed a cybersecurity club and created a secure study archive containing notes, practice problems, and exam hints. To protect the archive, they relied on cryptographic techniques learned in the course and left behind encrypted clues instead of recorded access credentials.

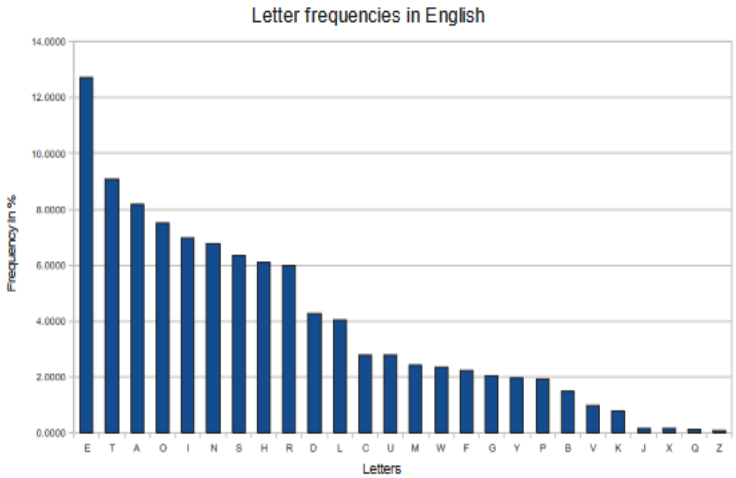
Your task is to reconstruct access information by analyzing and breaking these clues. You must determine the access date, access location (server name), and the final access code required to unlock the archive.

Important: Some decrypted outputs represent key codes or identifiers, not natural-language messages i.e. they may not make sense in English. The decrypted text does not need to be meaningful, as long as the cryptographic steps are correct and clearly shown.

Clue 1: The Oldest Note 10 Points

Ciphertext: QLL QXL QLLQX

Break the cipher using frequency analysis. Recover the plaintext and briefly explain why frequency analysis is effective against substitution ciphers. You can use the following frequency table.



19/36 Chapter 1 of *Understanding Cryptography* by Christof Paar and Jan Pelzl

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1) QLL QXL QLLQX

Q - A?
L - L?
X - H?

ALL AHL ALLAH

Due to the very short ciphertext, there are many valid substitutions that fit and make sense. I believe this is the best possible solution to the cipher as it produces at valid 5 letter word with the required pattern.

Clue 2: The Shifted Date 10 Points

Ciphertext: HLJKW
Key: k = 3

Decrypt the message using the Caesar cipher. Show the numeric mapping (A = 0, …, Z = 25) and interpret the result as a date-related clue.

2) k=3

H -> 7 - 3 -> 4 = E
L -> 11 - 3 -> 8 = I
J -> 9 - 3 -> 6 = G
K -> 10 - 3 -> 7 = H
W -> 22 - 3 -> 19 = T

PT = Eight

Clue 3: The Server Name (Affine Cipher) 15 Points

Ciphertext: RCLLA

Encryption rule: E(x) = (7x + 3) mod 26

Determine whether the cipher is invertible. Compute the modular inverse of 7 (if it exists) and decrypt the ciphertext to obtain the server name.

3) RCLLA

$E(x) = (7x+3) \% 26$

invertible if $\gcd(7, 26) = 1$

mod inverse of 7

$7 \times 15 = 105 \equiv 1 \bmod 26$

Decryption -> $15(x-3) \bmod 26$

R: 17 -> $15(17 \underset{14}{-} 3) \% 26$ -> $210 \% 26 \equiv 2$ -> C

C: 2 -> $15(2 \underset{-1}{-} 3) \% 26$ -> $-15 \% 26 \equiv 11$ -> L

L: 11 -> $15(11 \underset{8}{-} 3) \% 26$ -> $120 \% 26 \equiv 16$ -> Q

L:

A: 0 -> $15(0 \underset{-3}{-} 3) \% 26$ -> $-45 \% 26 \equiv 7$ -> H

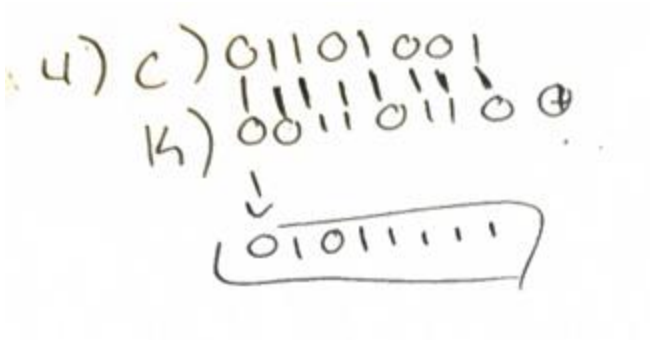
CLQQH

c) Factor invali & 13 valid #

Clue 4: The Streaming Key (Stream Cipher) 10 Points

Ciphertext (C): 01101001
Keystream (K): 00110110

Recover the plaintext using XOR. Explain why XOR allows both encryption and decryption, and explain why reusing the same keystream would compromise security.



XOR allows for both encryption and decryption because it ahas a reversible property that can be utilized, and applying the same key twice restores the original plaintext. Reusing a keystream would compromise security because XORing 2 ciphertexts removs the keystream.

Clue 5: The Document Passcode 20 Points

After working through all of the earlier challenges, you decide to join the group of students in the cybersecurity club. As a new member, you are tasked with protecting a shared study document that includes notes, practice problems, and exam hints. To ensure the document remains secure, you design a passcode using the principles of confusion and diffusion, similar to how modern block ciphers are designed.

You start with the following 8-bit base passcode, shown in 2-bit blocks for clarity:
Initial passcode (P):
10 11 00 01

Confusion (Substitution)

To make the passcode harder to guess, you first apply a substitution rule to each 2-bit block using the table below. This step hides the relationship between the original passcode and the final result.
Apply this substitution independently to each 2-bit block of the passcode.

Input	Output
00	11
01	10
10	01
11	00

Step 2: Diffusion (Permutation)

Next, you spread the influence of each bit by rearranging the bits of the substituted passcode using the following permutation rule:

Output bit positions: 3 6 1 8 5 2 7 4

a. Apply the substitution step to each 2-bit block of the initial passcode.

5) 10 11 00 01

I	O
00	11
01	10
10	01
11	00

1 2 3 4 5 6 7 8
01 00 11 10

b. Combine the substituted blocks and apply the permutation to generate the final secure passcode.

b) Permutation
3 6 1 8 5 2 7 4
01001110

c. Flip one bit in the original passcode and repeat steps (a) and (b).

c) 10 11 00 01

00 11 00 01 Flipped 1st bit
1 2 3 4 5 6 7 8
11 00 11 10

01101110

d. Compare the two final passcodes and describe what changes you observe. Explain how this demonstrates diffusion.

d) 01001110 vs 01101110
This shows a small difference by using the "flip 1 bit" rule, resulting in a similar output. The avalanche effect is found in strong diffusion, which results in changing many output bits. Real ciphers do this by using multiple rounds.

e. Defend your design. In your own words, explain why using both confusion and diffusion makes the passcode more secure than using only substitution or only permutation. Explain how each step contributes to protecting the club's document.

e)
Confusion: Hides the relationship between original bits & the result.
Diffusion: Spreads where each bit ends on the final output rather than staying in the original 2 block positions.
Combining both forces changes that make it very hard for attackers to predict & recognize patterns that would easily be decrypted

Clue 6: The Algorithm They Rejected (DES) 10 Points

Identify two limitations of DES.

1) Small key size of 56 bits
2) 64 bit block size is small for todays standards

Explain why DES is no longer considered secure in practice and briefly explain why DES remains important as a teaching cipher.

DES is no longer secure as it can be brute forces and an attacker can keep breaking the ciphertext until it looks correct as PT.

As a teaching cipher however, its an easily understood example of a Feistel network and shows the core ideas of block ciphers. Understanding DES allows students to understand how modern designs work even though DES itself shouldn't be used anymore, the core ideas and fundamentals are very valuable.

Clue 7: The Final Access Code 20 Points

The final access code is constructed using properties of modular arithmetic.

a. First digit: The number of elements that have a multiplicative inverse in the ring of 8 ie Z_8.

7) a) $Z_8 = 0, 1, 2, 3, 4, 5, 6, 7$

• gcd(x, 8)

(0, 8) ⇒ 8 — 4 elements

(1, 8) = 1

(2, 8) = 2 — 1st digit = 4

(3, 8) = 1

(4, 8) = 4

(5, 8) = 1

(6, 8) = 2

(7, 8) = 1

~~(8, 8)~~

b. Next two digits: The number of elements that have a multiplicative inverse in the ring of 34 i.e. Z_{34}.

b) Z_{34}

• Multiples of 2 in 0-33

↓

17 numbers

0, 17 = 17+1 = 18

34 − 18 = 16

16

c. Next digit: The number of valid keys for an Affine cipher modulo 26.

c) $E(x) = ax + b \bmod 26$

Factors of 26 → 2×13,

invalid # = Even # from 0-25 & 13

valid # All odd from 0-25

12 valid "a"

26 valid "b"

$12 \times 26 = \boxed{312}$

d. Last four digits: Add together all digits of your L number and take the result modulo (the last digit of your student ID plus one). Write the result as a four-digit number, including leading zeros if necessary.

d) L number: 00308579

3 + 11 + 16 = 32

32 mod (9+1)

32 mod 10 = 2

0002

Final Access Code:

4163120002

Show all calculations clearly and report the final access code.

Final Reflection 5 Points

In 3–5 sentences, explain how cryptography evolved in this course from substitution ciphers, to stream ciphers, to block ciphers using confusion, diffusion, and multiple rounds. Explain why layered cipher design is essential for modern cryptographic security.

Cryptography progressed from simple substitution ciphers all the way to advanced stream and block ciphers. Earlier ciphers show how predictable patterns and freqencies can make encryptions vulnerable to attacks, while stream ciphers have introduced key streams and XOR transformations. Block ciphers evolved to show how confusion and diffusion work together with multiple rounds to resist/delay attakcs. The layered design is critical as modern security relies heavily on making relationships between the plaintext, ciphertext, and keys very difficult to analyze or exploit.